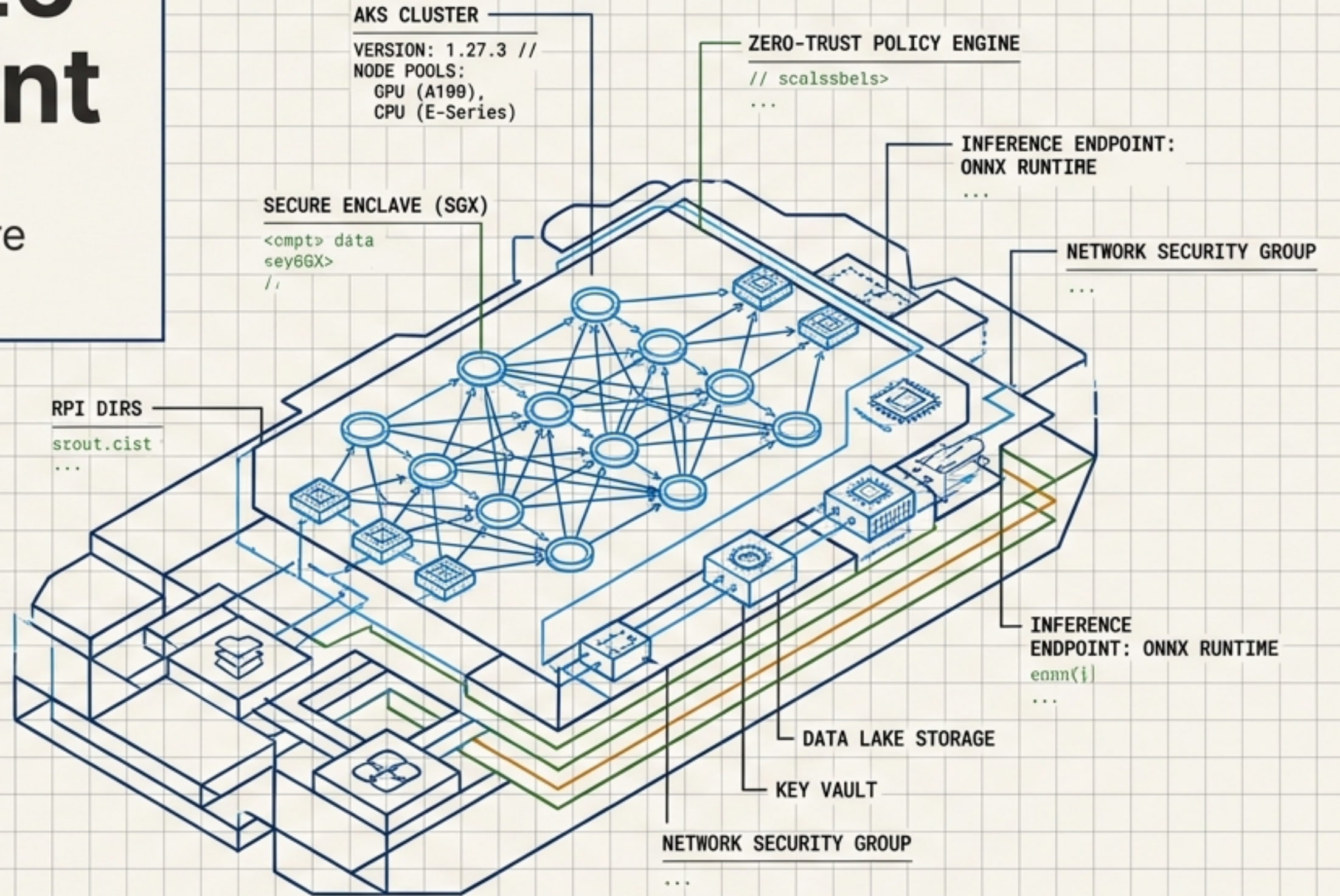


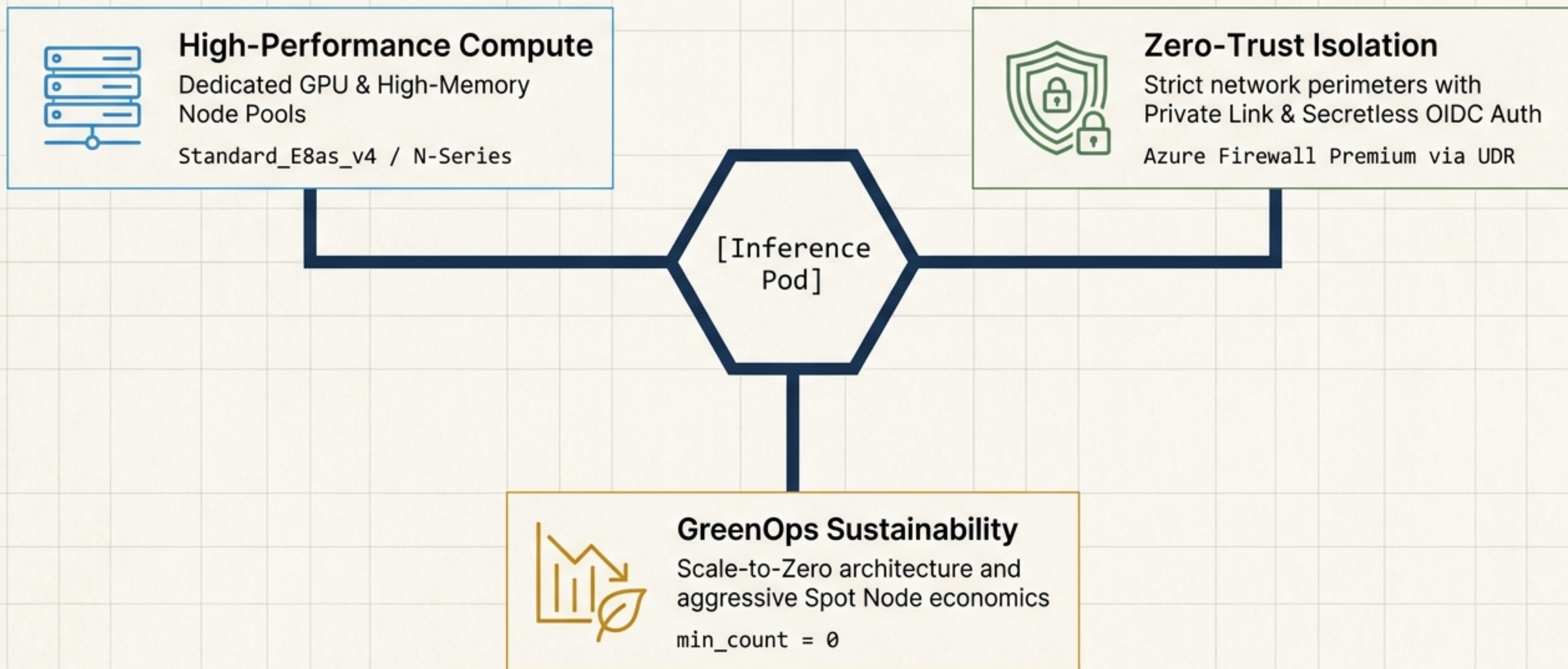
The Vision 2026 MLOps Blueprint

Architecting Secure, Zero-Trust
Machine Learning Inference on Azure
Kubernetes Service (AKS)



STATUS: PRODUCTION-READY // IaC SOVEREIGNTY: ACTIVE

Executive Summary: The MLOps Ecosystem



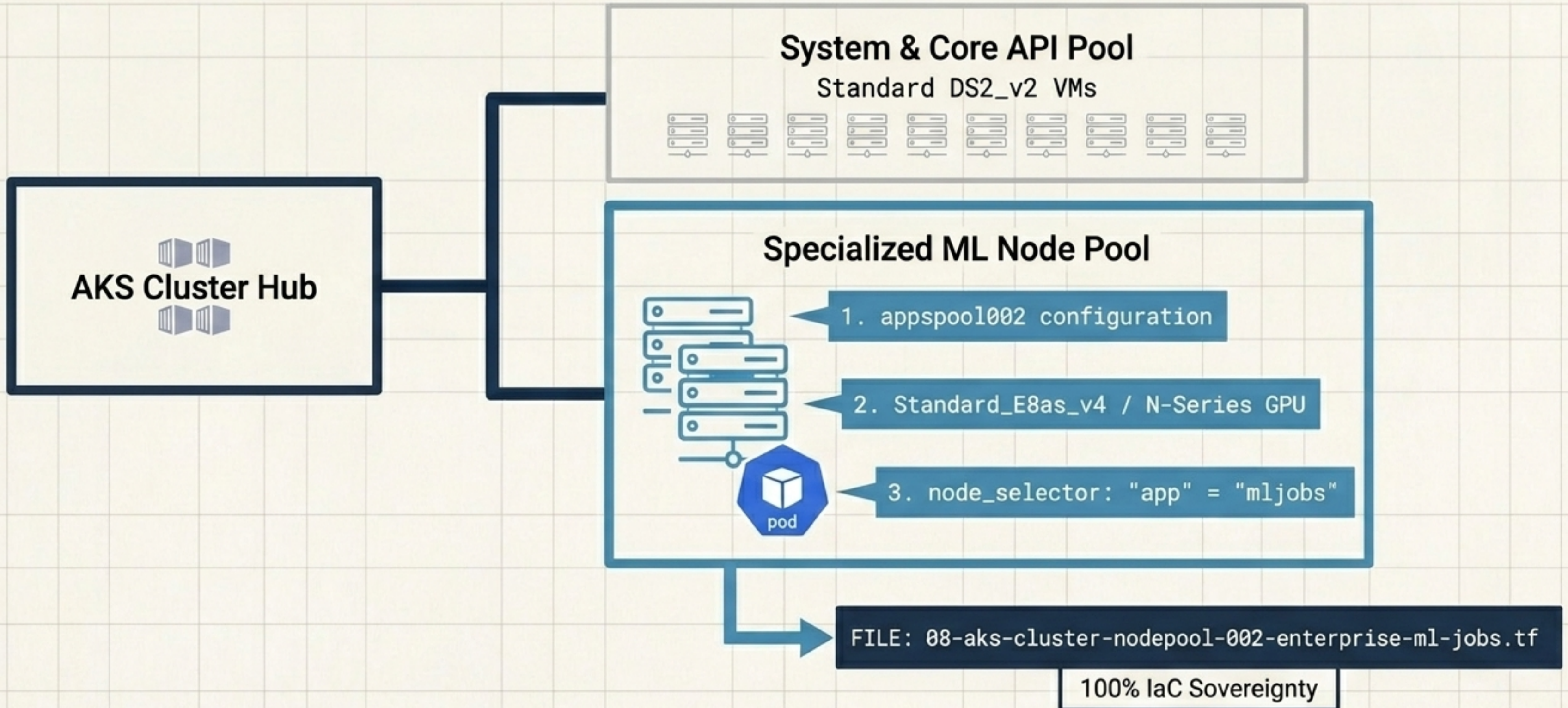
Architectural Strategy: Why Isolate ML Workloads?

Workload Contention

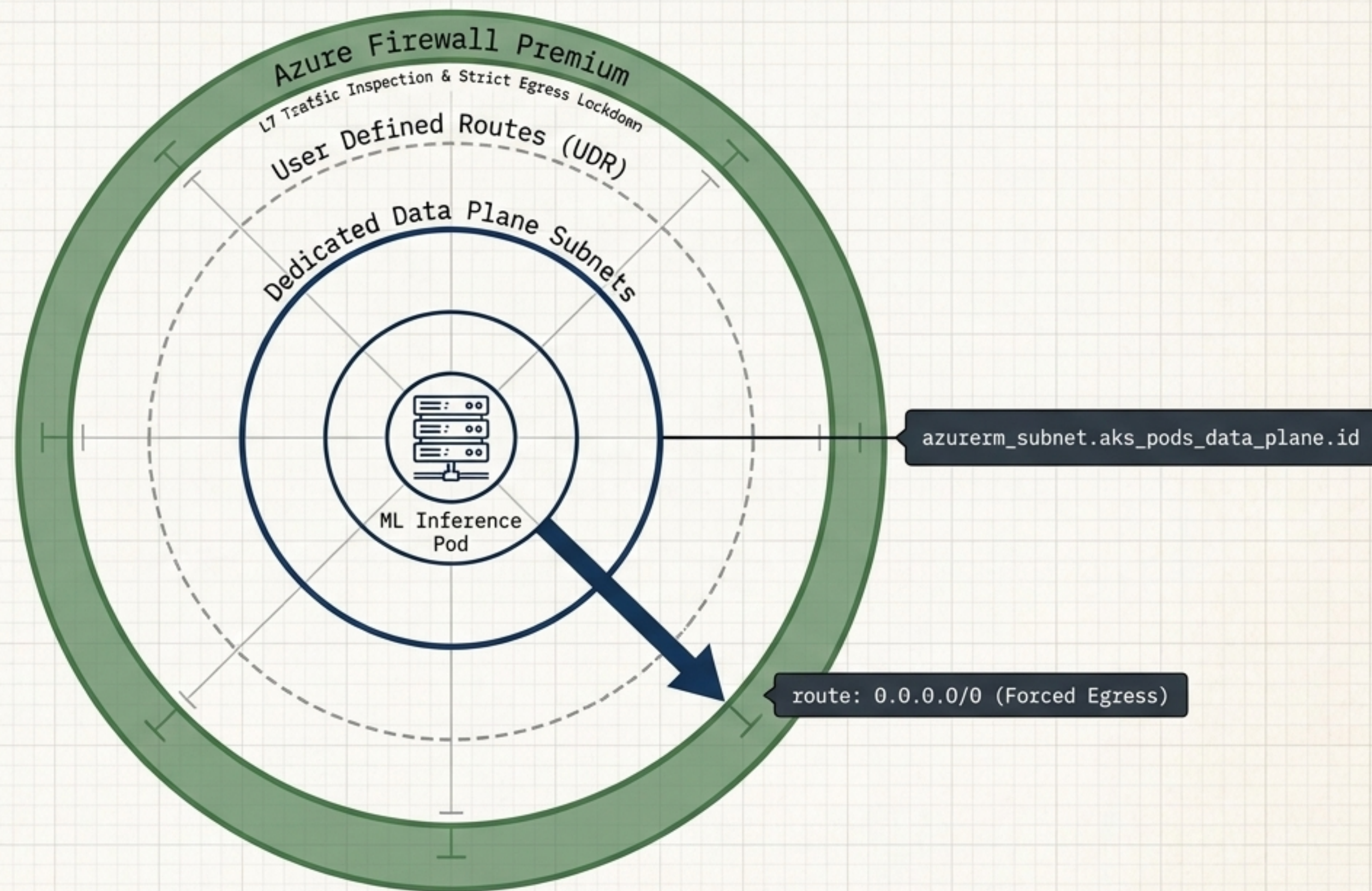
Core Business APIs	ML Inference Workloads
Compute Need: Standard, uniform CPU allocation.	Compute Need: Extreme burst High-Memory/GPU. (N-Series, Standard_E8as_v4)
Scaling Pattern: Predictable, linear web traffic. 	Scaling Pattern: Highly volatile, sporadic inference bursts. 
Security Profile: Standard database access (OBO flow).	Security Profile: Multi-GB proprietary model weights; high intellectual property risk.

TAKEAWAY: Running ML workloads on general compute causes resource contention and core limit exhaustion. Specialized node pool isolation is mandatory.

Pillar 1: High-Performance Compute Isolation

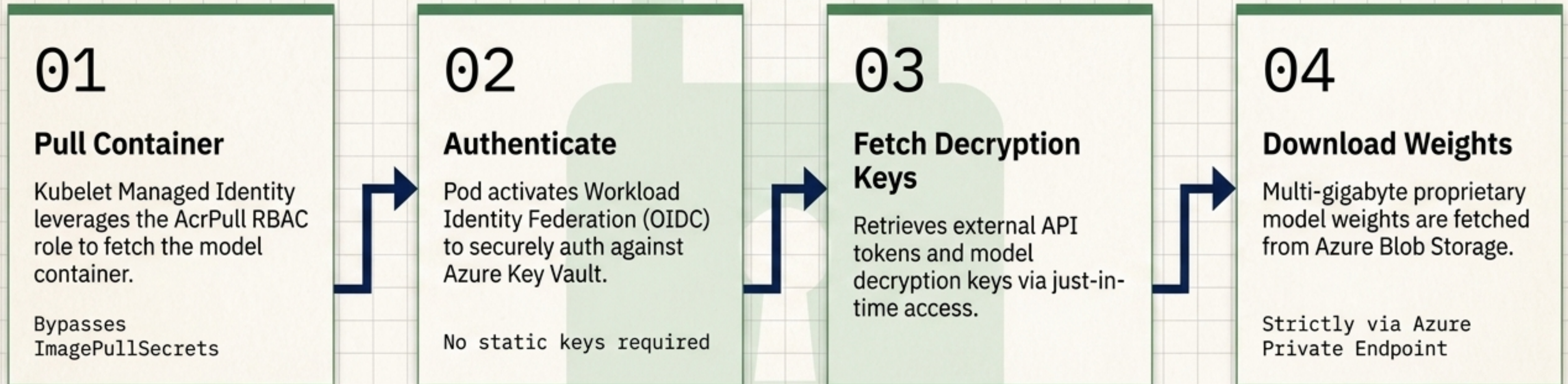


Pillar 2: The Zero-Trust Security Fabric



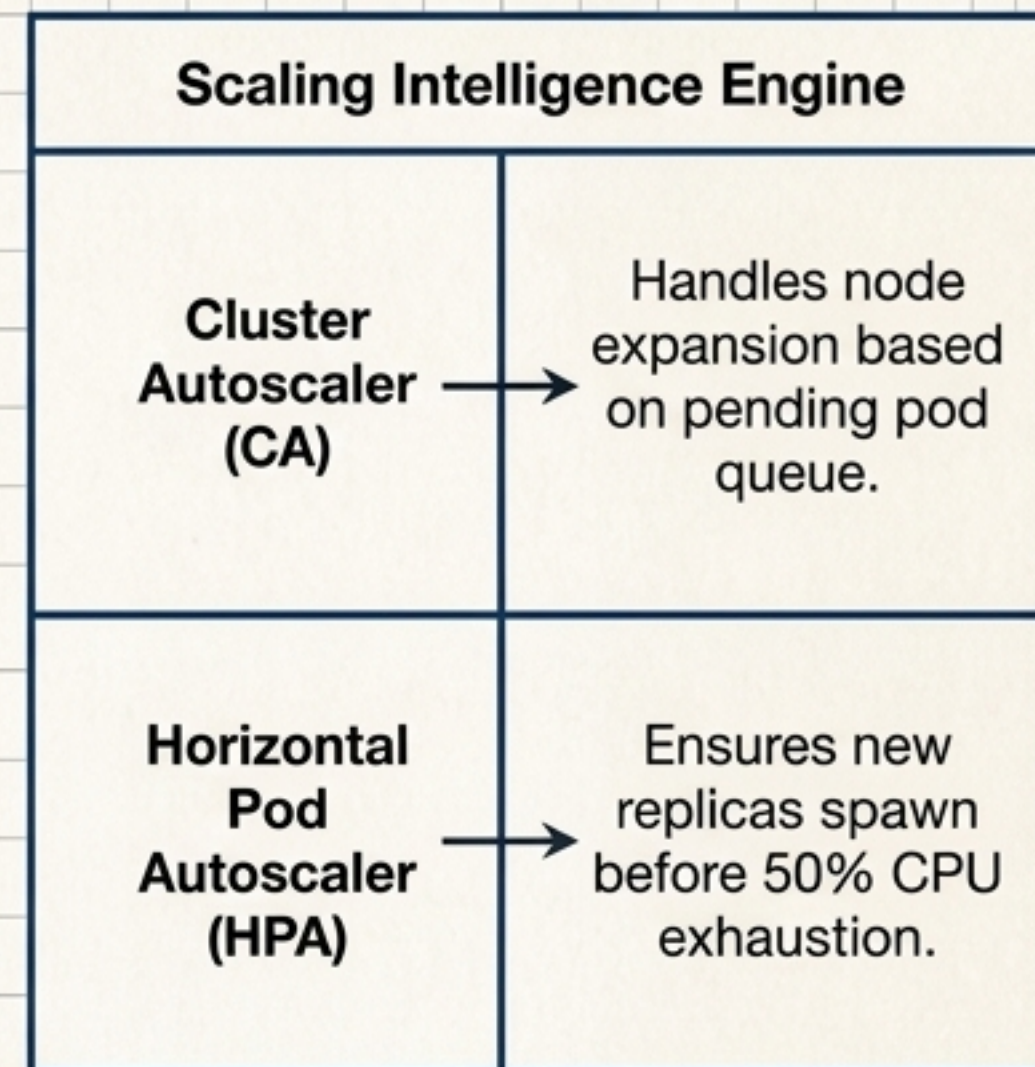
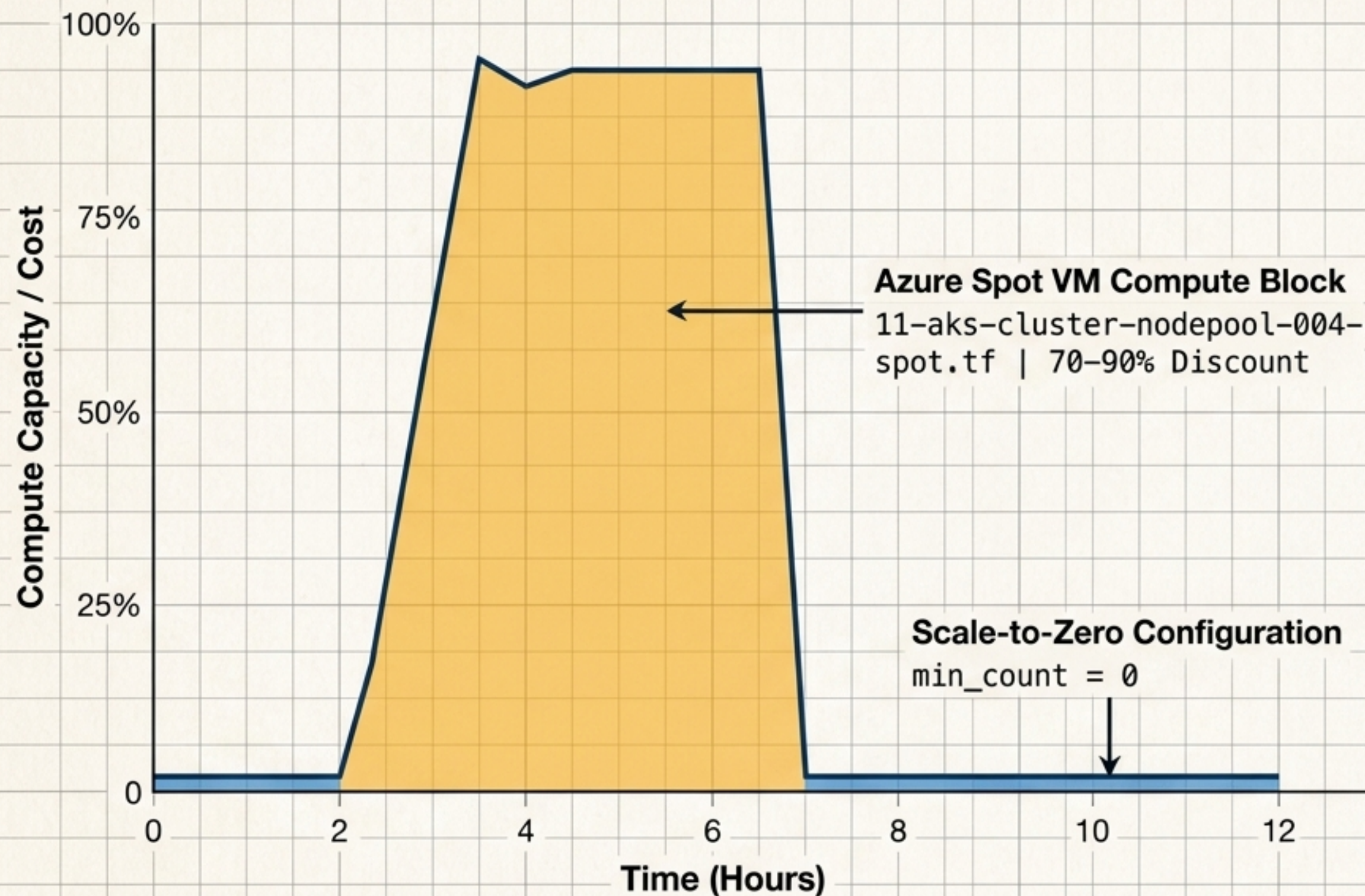
PUBLIC INTERNET
ACCESS: BLOCKED

Step-by-Step: The Zero-Secret Model Delivery Flow



CRITICAL OUTCOME: The public internet is entirely bypassed during model initialization.

Pillar 3: GreenOps & Scaling Economics



The IaC Foundation: Deploying the ML Stack

.tfvars environment mapping

Phase 1: Shared-Infra

Foundational Hub-Spoke networking,
Private DNS, Firewall

Phase 2: AKS Compute

Managed clusters, ML node pools,
OIDC issuer setup

Extracts Triad certificates to
authenticate Helm providers

Phase 3: Day2 Ops

Prometheus, Grafana, KubeCost for
MLOps observability

Immutable Deployment Execution

100% Terraform
driven. Manual
portal changes
are identified as
drift and
automatically
destroyed to
enforce
configuration
sovereignty.

Architecture Wiring: Shadow Infrastructure Naming Logic

The Input

```
git checkout branch develop
```

03-locals.tf
Engine

```
gitbranch = (var.gitbranch != 'main') ? 'd':''
```

The Output

Branch = develop

Prepends 'd'
to all resources.

rg-dnede

Shadow Infrastructure.
Total isolation for experimental ML node testing.

Branch = main

Clean naming,
no prefix.

rg-nepro

Mission-Critical.
Immutable, gold-standard infrastructure.

Advanced Constraints: Referential Attribute Linking

Hardcoded Strings & Lookups



```
subnet_id =  
  '/subscriptions/123/resourceGroups/...'
```

Diagnostic Notes:

Causes Chicken-and-Egg dependency failures.
Breaks when ML subnets are rapidly recreated
during rapid iteration.

Referential Attribute Linking



```
resource 'azurerm_subnet' 'aks_nodes_data_plane'
```

```
vnet_subnet_id =  
  azurerm_subnet.aks_nodes_data_plane.id
```

Diagnostic Notes:

laC Sovereignty.
Creates an implicit dependency graph.
Terraform natively understands execution order.

Synthesis: The MLOps Risk vs. Resolution Matrix

Enterprise ML Risk	Vision 2026 Solution
Model Theft & Intellectual Property Exfiltration	Forced Tunneling (UDR) + Private Link for Blob Storage and Key Vault.
Long-lived Credential Leaks in CI/CD	OIDC Federation + On-Behalf-Of (OBO) Flow for Zero-Secret architecture.
Compute Contention (Core APIs crash under AI load)	Dedicated Node Pools with strict pod scheduling labels (app=mljobs).
Runaway AI Infrastructure Costs	Scale-to-Zero logic (min_count = 0) seamlessly integrated with Spot VMs.

SECURE, IDEMPOTENT, AND INFINITELY SCALABLE MLOps – 100% DEFINED IN CODE.